

$$(0,0,1), (1,0,1), (0,1,1), (1,1,2), (1,-1,0), (-1,1,1)$$

$$z = ax + by + c \quad \text{t.j.} \quad \sum_i (z_i - (ax_i + by_i + c))^2 \text{ minimal!}$$

Podle 8.91 hledáme $\hat{x}$ t.j. $A^T A \hat{x} = A^T \vec{b}$

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} \quad \vec{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} \quad \hat{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$\begin{matrix} \text{absolutní} \\ \text{člen } c \end{matrix}$
 $\begin{matrix} \text{1-ové} \\ \text{souřadnice} \end{matrix}$
 $\begin{matrix} \text{2-ové} \\ \text{souřadnice} \end{matrix}$

$$A^T A = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} \quad A^T \vec{b} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

Dostáváme SLR:

$$\left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{array} \right) \quad \text{odkud máme řešení} \quad \hat{x} = \frac{1}{5} \begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix}$$

Nově hledané roviny je tedy $z = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}$